The University of the South Pacific

Serving the Cook Islands, Fiji, Kiribati, Marshall Islands, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, and Vanuatu.

School of Computing, Information and Mathematical Sciences

CS211: Computer Organisation

Test 2 – Semester 2, 2014

Time Allowed: 1 hour

Total Marks: 30 (15% of final grade)

Name: Solution **Student ID:** _____

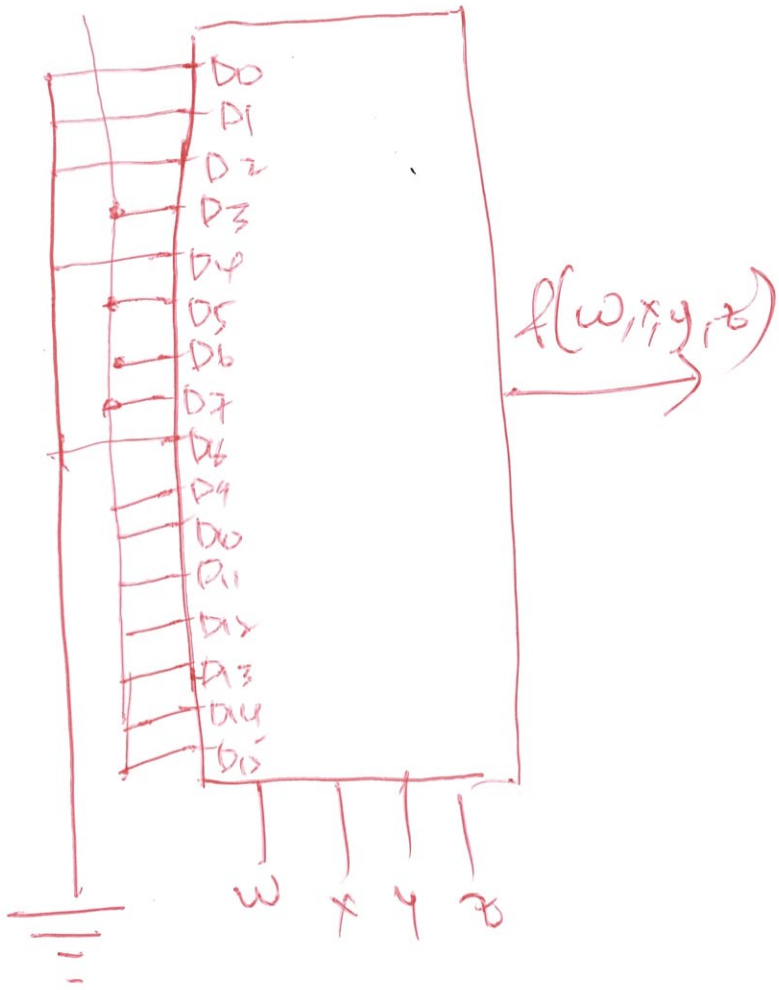
INSTRUCTIONS:

- There are 6 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

Using truth table draw the four variable multiplexer chip for the function whose output is the most common value among four inputs. In case of a tie, output 1. (3 marks)

w	x	y	z	f(w,x,y,z)
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1



Question 2

1) Assume the following instructions are available for 0-address (stack-based) and 1-address machines: ADD, SUB, MUL, DIV, LOAD, STORE, PUSH, POP. Memory addresses like A, B, C, etc. are used as operands. Each instruction is made up of 24 bits operand and 8 bits of opcode.

   a) Write instructions for a 0-address (stack) machine to compute X = (A-B)/(C+D/E). (2 marks)

   PUSH A
   PUSH B
   SUB
   PUSH D
   PUSH E
   DIV
   PUSH C
   DIV
   POP X

   b) How many bits does 0-address machine need to compute X? (1 mark)

   216

   c) Write instructions for a 1-address machine to compute X = (A-B)/(C+D/E). (2 marks)

   LOAD D
   DIV E
   ADD C
   STORE T
   LOAD A
   SUB B
   DIV T
   STORE X

d) How many bits does 1-address machine need to compute X? (1 mark)

256

Question 3

Design an expanding opcode to allow the following to be encoded in a 22-bit instruction with each address being 10bits. (5 marks)

3 two-address instructions
24 one-address instructions
15 zero-address instructions

00	AAAAAAAAAA	BBBBBBBBBB	3
01	"	"	
10	"	"	
11	0000000000	BBBBBBBBBB	24
	⋮		
11	0000010111	"	
11	0000011000	0000000000	15
⋮	⋮	⋮	
11	0000011000	0000001110	

Question 4

Use Boolean algebra to show that: $xz = (x + y)(x + \bar{y})(\bar{x} + z)$. (4 marks)

$(x+y)(x+\bar{y})(\bar{x}+z)$

$(xx + yx + x\bar{y} + y\bar{y})(\bar{x}+z)$

$(x + xy + x\bar{y} + 0)(\bar{x}+z)$

$(x + x(\cancel{y+\bar{y}}) + 0)(\bar{x}+z)$

$(x + x)(\bar{x}+z)$

$x(\bar{x}+z)$

$\cancel{x\bar{x}} + xz \Rightarrow \underline{\underline{xz}}.$

Question 5

2) Assume that you have a byte-addressable machine that uses 32-bit word. Show how the hexadecimal value ABAB will be stored at address 0 if:

a) the machine is big endian? (2 marks)

0	00000000	00000000	1010 1011	10101011

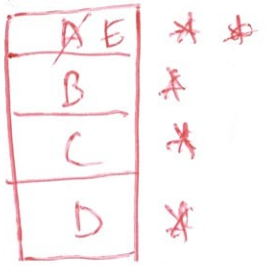
b) the machine is little endian?(2 marks)

	00000000	00000000	1010 1011	1010 1011	0

Question 6

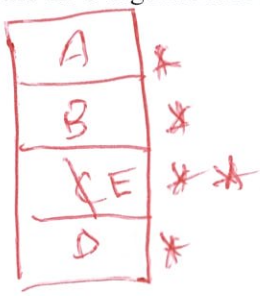
Assume that memory has 4 page frames, 5 virtual pages (A-E) and the following page reference sequence occurs: A, A, A, B, A, B, C, D, D, D, E

a) If the LRU algorithm is used, which page will be replaced? (2 marks)



A will be removed.

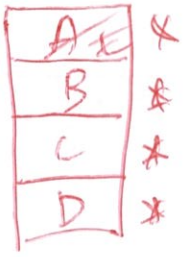
b) If the LFU algorithm is used, which page will be replaced? (2 marks)



A = 4
B = 2
C = 1
D = 3

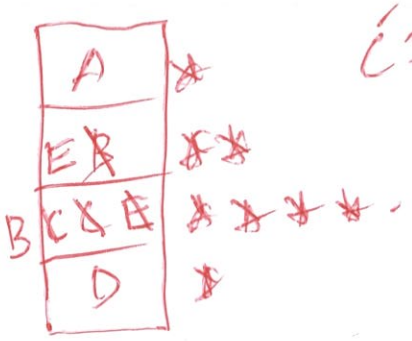
C will be removed.

c) If the FIFO algorithm is used, which page will be replaced? (2 marks)



A will be replaced.

d) If the LFU algorithm is used, give 3 page references after page E that will cause a page fault? (2 marks)



CEB

The End

[EXTRA PAGE]